

Supply Chain Exception Control System

Logical Data Model and Data Contracts

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Version: 1.0

Status: Frozen for physical PostgreSQL design, subject only to the open decisions in Section 15 and Appendix G

Classification: Public portfolio-safe when populated only with synthetic data

Control position. The model separates analytical assessment from operational case management. A candidate risk evaluation is immutable analytical evidence. An exception episode is the human-controlled operational record. Closed is the only terminal state. Suppressed and Resolved remain active for uniqueness and audit purposes. This document defines logical contracts; it does not claim production readiness.

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Appendices A-H provide the recommended entity list, grain and cardinality summaries, uniqueness and data-quality contracts, decision register, pre-build questions and critical self-review.

Document authority and conflict handling

This specification is governed by the Project Charter v1.0, Business Requirements Specification (BRS) v1.0, and Frozen Operational Exception Lifecycle and Operating Model v1.0. Where the BRS lifecycle wording conflicts with the later frozen operating model, the frozen operating model governs.

The following supersessions are material:

1. BRS LIF-007 allows limited automatic resolution; Frozen LIF-DEC-003 and AUT-DEC-001 prohibit automated Resolved and Closed transitions in the MVP. Automation may record possible-resolution evidence only.
2. BRS LIF-010 says reopening may reset an SLA; Frozen SLA-DEC-005 prohibits a same-episode reset. A post-Closed recurrence creates a successor episode with new clocks.
3. The BRS transition matrix includes Candidate Risk to Suppressed. Frozen LIF-DEC-002 makes Candidate Risk analytical and LIF-DEC-005 makes Suppressed an operational episode state. Duplicate/cooldown candidates therefore receive a candidate disposition and, where relevant, link to an existing episode; they do not become lifecycle states.
4. The Charter discusses SQLite-first delivery. The confirmed baseline and BRS NFR-009 make PostgreSQL the deployed source of truth; SQLite is limited to local tests.

Source references in this document use document name plus requirement or decision ID, for example “BRS FR-007” and “Operating Model DUP-DEC-001.”

1. Data architecture overview

1.1 Architectural principles

- **Grain first.** The operational grain is one purchase-order-line risk episode for one stable line identity at one receiving site.
- **Immutable observations, controlled projections.** Source observations, score evaluations, contributions, workflow events, approvals, receipts, and corrections are retained. Current status, owner, risk, and balances are projections.
- **One authority per concept.** Raw data proves what arrived; validated operational entities establish accepted business facts; workflow events establish what users and automation did; curated analytics establishes report-ready derivations but never becomes an operational write-back source.
- **No silent repair.** Corrections append a new record linked to the superseded record. Reversals neutralise effects but preserve both records.
- **Version everything that changes interpretation.** Risk rules, business calendars, ownership mappings, master attributes, UOM conversions, and reporting publication identities are effective-dated or run-versioned.
- **Business time is explicit.** Instants are stored timezone-aware. Australia/Melbourne local date, local time, and business-minute calculations are derived using the calendar version that applied to the event or SLA obligation.
- **Independent outcomes.** Synthetic outcome labels and their seeds/configuration are isolated from risk-rule versions and scoring inputs. [BRS DQ-010]

1.2 Logical layers and authorities

Layer	Purpose	Representative content	Authoritative for	Not authoritative for
Source/raw	Preserve each delivered extract without business reinterpretation	Source file/object metadata, row payload/hash, extraction time, source-system key	What the project received and when	Valid business facts, current workflow state
Staging and validation	Parse, standardise, validate, reconcile and disposition every row	Typed staging rows, validation results, quarantine/rejection, control totals	Acceptance/rejection decision for a run	Long-lived master or transaction identity
Reference/master data	Establish stable identities and versioned policies	Site, supplier, product, user, UOM, product-site policy, ownership mapping, rule/calendar versions	Canonical identities and effective-dated business configuration	Source transactions or case history
Operational transactions	Record accepted procurement and supply observations	PO, line, schedule, commitment, receipt, receipt allocation, inventory, demand	Accepted procurement facts and their correction history	Risk judgement or lifecycle state

Layer	Purpose	Representative content	Authoritative for	Not authoritative for
Candidate and scoring	Evaluate each eligible line once per successful scoring run	Evaluation, contribution, missing signal, severity, disposition	Historical calculated risk for a run and rule version	Human decision, current case status, realised outcome
Exception workflow	Manage the risk episode and its event history	Episode, state event, action, ownership event, SLA event	Operational case identity and controlled event history	Source truth or score probability
Approvals and evidence	Prove authority and supporting basis	Approval request/decision, suppression control, evidence reference/link, resolution record	Decision authorisation and evidence metadata	Binary document storage, enterprise records management
Notifications	Record intended and attempted communications	Notification event, channel, recipient reference, delivery outcome	Notification attempt/delivery history	Escalation itself or workflow state
Pipeline control	Control ingestion, validation, scoring and publication	Pipeline run, source load, step result, reconciliation, current-publication pointer	Run status, lineage, freshness and safe publication	Operational case state
Analytics/reporting	Provide reconciled current and historical read models	Facts, dimensions, snapshots, KPI definitions, publication manifest	Report-ready derived measures tied to a successful publication	Operational write-back or mutable source facts

1.3 Authoritative concept map

- Stable PO-line identity: **purchase-order line identity and key-alias history**.
- Current accepted order facts: **latest non-reversed PO/line/schedule versions**, never the raw layer.
- Receipt history: **receipt transaction**; receipt-to-schedule application: **receipt allocation**.
- Current residual quantity: derived from accepted line/schedules minus effective allocations, bounded by correction/reversal rules.
- Current calculated risk: latest eligible **candidate risk evaluation** from the latest successful published scoring run for the line/site.
- Current operational severity: calculated severity unless a valid approved time-bound override applies; both values remain visible.
- Current exception state: latest valid **exception state event** ordered by effective time then recorded sequence.
- Current owner: latest effective **ownership event**, considering approved delegation intervals.
- SLA condition: derived from SLA obligation/pause intervals and immutable breach/escalation events under the recorded calendar version.
- Approval authority: **approval decision**, never a free-text action note.
- Resolution: **resolution record plus state event**; possible-resolution evidence alone has no state effect.
- Reporting currency: latest successful **analytics publication**, not the latest attempted run.

2. Logical entity catalogue

2.1 Source, control and validation

Source System

Registry for each simulated ERP, WMS, inventory, demand or supplier-performance source. Grain: one source-system identity. Relates one-to-many to source loads, business-key aliases and raw records. Version connection/display metadata, but never store credentials.

Source Load

Added because a pipeline run may ingest several datasets with different extraction times and row counts. Grain: one delivered dataset/object in one pipeline run. Records dataset type, source, URI-safe reference, content hash, schema version, extracted-at, received-at, row counts and freshness. This entity is the primary lineage anchor for accepted and rejected rows.

Pipeline Run

One immutable processing attempt, including failed attempts. Records run type, trigger, status, Melbourne and UTC timing, code/configuration release, rule/calendar versions where applicable, counts and publication eligibility. A run may contain many source loads and candidate evaluations. [BRS FR-002, DQ-011, REP-007]

Pipeline Step Result

Recommended generic control entity rather than separate ingestion, validation, scoring and publication run tables. Grain: one named step attempt within a pipeline run. It records start/end, status, counts, error classification and input/output hashes.

Rejected Record

Consolidates reject and quarantine metadata into one entity with disposition. Grain: one validation failure for one source row and rule. Multiple failures may exist for one row. Stores defect code, field, safe observed value/hash, severity, disposition (reject row, quarantine row, stop dataset, stop run, warning), correction linkage and resolution status. Raw confidential values are unnecessary in the public portfolio.

2.2 Reference and master data

Site

One receiving/operating site identity. Versioned attributes include name, state, timezone and active interval. Both simulated sites use Australia/Melbourne business semantics in the MVP.

Supplier and Supplier Version

Supplier provides the durable identity; Supplier Version contains effective-dated descriptive and classification attributes. This separation prevents historical evaluations from inheriting later supplier categories or names. Supplier performance is a separate observation, not an attribute overwrite.

Product and Product Version

Product provides durable SKU identity; Product Version contains effective-dated description, category, base UOM, handling precision and lifecycle attributes.

Product-Site Inventory Policy

Grain: one product/site policy version per effective interval. Contains safety stock, reorder/coverage assumptions, allowed substitution group and policy source. Overlapping effective intervals for the same product/site are prohibited.

User

One simulated employee/system actor identity. Contains synthetic display name, role classification, active interval and actor type. Authentication secrets and real personal identifiers are excluded.

Ownership Mapping

One effective-dated assignment rule at a stated precedence and scope: supplier-site exception, buyer-group/site, category/site, duty queue. It points to an owner or queue, has approval/change evidence and must not overlap another mapping at the same precedence and exact scope. [Operating Model OWN-DEC-001]

Rule Version

One approved immutable scoring-rule package version. It includes effective interval, owner, status, rationale, approval and test-evidence references. Component rules may be represented as child rule definitions; an activated version cannot be edited.

Calendar Version

One immutable business-calendar version containing timezone, business hours and holiday set. The frozen MVP baseline is Monday-Friday 08:30-17:00, Victorian holidays, Australia/Melbourne DST. An SLA obligation retains the calendar version with which it was created. [Operating Model SLA-DEC-001]

UOM Conversion

Added as a small reference entity because receipt and exposure quantities cannot be safely compared without it. Grain: one product/from-UOM/to-base-UOM conversion version. The MVP should prefer exact fixed conversions; missing or ambiguous conversions quarantine affected calculations.

2.3 Procurement and supply observations

Purchase Order

Durable order-header identity within a source system. Header versions preserve supplier, buyer group, currency, order date and status changes. One order has one or more lines.

Purchase-Order Line Identity and Line Version

The identity provides a generated canonical key independent of mutable source attributes. Key aliases retain source PO/line references and corrections. Line versions record ordered quantity, UOM, receiving site, product, requested/need date, cancellation/status and amendment sequence. This split is necessary because source-key correction and genuine amendments have different semantics.

Delivery Schedule

Grain: one schedule identity/version for one PO line. Contains scheduled quantity, requested/confirmed/expected date, source schedule key and status. Split schedules remain components inside the PO-line episode and are not exception keys. [Operating Model EPI-DEC-002]

Supplier Commitment Observation

Added because BRS FR-001 requires confirmation data and the operating model distinguishes a promise from receipt. Grain: one supplier confirmation/commitment observation for one line or schedule at one observed time. It records committed quantity/date, channel, source load and whether it supersedes or corrects a prior commitment.

Receipt Transaction

One source receipt event or correction/reversal transaction. The original source receipt identity is stable; corrections append linked transactions. Quantity is retained in source and product base UOM where conversion is valid.

Receipt Allocation

Added because receipts may not identify a schedule and partial/split receipts must be reproducible. Grain: one application of one effective receipt transaction to one delivery schedule or explicit line-level residual bucket. Allocations never exceed the effective receipt quantity; overreceipt remains explicitly unallocated/excess.

Inventory Snapshot

One product/site inventory observation at a snapshot time and source load. Quantities include on-hand, allocated/reserved, available and optionally in-transit, each defined consistently. Later snapshots do not correct earlier snapshots; a correction is a new linked snapshot.

Demand Requirement

One dated product/site demand requirement from a named demand source. It may represent sales order, forecast bucket or planned requirement; requirement type is mandatory so double counting can be controlled. Corrections/reversals append.

Supplier Performance Snapshot

One supplier/site or supplier-level performance observation for a defined measurement window and as-of date. Contains numerator, denominator, sample sufficiency, OTIF and approved definition/version. It is not recomputed in place.

Synthetic Outcome Observation

Added to enforce independence from scoring. Grain: one hidden simulated outcome for one line/site outcome window and generator version/seed reference. It is accessible to evaluation/reporting tests, not scoring inputs. [BRS DQ-010]

2.4 Candidate and scoring

Candidate Risk Evaluation

One evaluation of one eligible PO-line/site in one scoring run under one rule version. It records input snapshot references, eligibility, score, calculated severity, explanations, missing signals, disposition and optional linked active episode. Exactly one accepted evaluation exists for each eligible line/site/run. It is immutable. [BRS LIF-001, FR-006 to FR-010]

Candidate Risk Contribution

One component rule result within one candidate evaluation. Stores rule/component version, evaluated availability, observed value or safe representation, threshold/comparator, trigger, points, explanation code and input lineage. Contributions must arithmetically reproduce the total score.

Candidate disposition is a controlled code, not a lifecycle state. Minimum values are: below-opening-threshold; opened-new-episode; linked-existing-active-episode; suppressed-by-existing-control; ineligible-after-validation; manual-review-data-insufficient; and scoring-error. "Suppressed-by-existing-control" means an effective episode-level suppression affected operational presentation; it does not create a Candidate state.

2.5 Exception workflow, approval and evidence

Exception Episode

One materially related risk episode for one stable PO-line/site identity from Open until Closed. It carries a generated episode ID, episode sequence, opening candidate/run, predecessor relationship and current projection pointers. It does not overwrite history. Resolved and Suppressed episodes remain active; only Closed is inactive for uniqueness.

Exception Event Envelope

Added as a common audit envelope for all controlled events. Grain: one recorded business event. Mandatory fields include episode, event type, effective-at, recorded-at, actor, actor type, reason/source, correlation, causation, before/after representation, pipeline/rule/calendar context and correction link. Typed child entities hold domain-specific fields. This avoids duplicating audit columns while preserving strong typed contracts.

Exception State Event

Typed event for one permitted lifecycle transition. Records from-state, to-state, transition reason, authority and any resolution/suppression reference. The first event is no-state to Open. Reopened is recorded as an event reason that returns a pre-Closed episode to Investigating or its configured active state; it is not a state.

Exception Action

One appended operational action, note, supplier contact, commitment, mitigation or plan amendment. Action records may be superseded but not edited; completion/cancellation is another event. Action Agreed requires a compliant plan record.

Ownership Event

One assignment, reassignment, delegation start/end or unassignment event. It stores previous/new owner, mapping/version basis, authority and effective interval. Reassignment neither changes lifecycle state nor resets SLA. [Operating Model OWN-DEC-003]

SLA Obligation and SLA Event

The requested “SLA event” is split logically. SLA Obligation defines each clock (assignment, action-plan, closure review, next action), original due time, calendar version and severity basis. SLA Event records start, pause, resume, extension, due, breach, escalation, satisfaction or cancellation. This separation prevents a mutable due date from erasing its origin. Breach and escalation do not change owner or lifecycle state.

Approval Request and Approval Decision

Kept separate. One request may require a manager decision and, for selected Critical routes, governance concurrence. The request records change type, original/requested values, reason, evidence, requester and expiry. Each decision records role, outcome, conditions, time and independence check. Requested changes take effect only when the required decision set is complete and valid.

Suppression Control

One approved suppression interval for one active episode. Records prior state, reason/scope, start/expiry, review, approval, recurrence criteria, score context and SLA consumption at pause. Extensions are new approval/control records. Expiry/revocation returns the same pre-Closed episode to an active state without resetting clocks.

Evidence Reference and Evidence Link

Evidence Reference stores metadata only: synthetic type, label, source/actor, recorded time, integrity hash/checksum where appropriate, resolvable portfolio-safe reference, availability and correction status. Evidence Link is a generic bridge from evidence to events, actions, approvals, resolution, receipts or source corrections. It avoids putting many nullable foreign keys on evidence.

Resolution Record

One proposed/accepted resolution assertion for an episode, with structured category, cause, evidence, resolver, residual risk, outcome quantity/date, monitoring result, approval and current score reference. Withdrawal is an appended event; the original resolution is retained. A Resolved transition is human-controlled and references one valid resolution record.

Notification Event

One intended delivery attempt for one trigger and recipient reference. Contains channel, template/version, payload hash, created/sent/delivered/failed times and retry/correlation data. Notifications are deferred in the core MVP and never substitute for queue, breach or escalation events. [BRS FR-027]

Episode Relationship

Recommended over a dedicated predecessor/successor entity. Grain: one directional link between two episodes with relationship type such as material-recurrence, site-transfer, source-key-correction-merge or duplicate-consolidation. For normal recurrence, a successor has at most one direct predecessor, while one predecessor may have multiple later successors only if distinct site exposures justify it. Self-links and cycles are prohibited.

3. Grain definition

3.1 Common fact contract

Every fact-like record has a generated immutable identifier, source/run lineage where applicable, an effective business instant, a recorded instant, actor/system attribution, and a correction/reversal relationship when applicable. `effective_at` describes when the fact is true in business terms; `recorded_at` describes when the system learned or recorded it. Both are timezone-aware instants. Melbourne-local date/time is derived with the relevant calendar/timezone rules.

Normal retention for the portfolio is **no automated purge**. Retain the complete 24-month reference period, all acceptance runs, and all workflow/audit evidence for the life of the project. A real organisational retention period remains a pre-production policy decision; no physical deletion mechanism should be implemented before it is approved. [BRS GOV-009]

3.2 Entity grain matrix — identity and lineage

Fact-like entity	Exact grain	Natural business key	Generated ID	Primary lineage / effective time
Source load	Dataset object in one run	source + dataset type + source object ID/extract version	source_load_id	pipeline run; extracted_at
Pipeline run	Processing attempt	orchestration attempt reference	pipeline_run_id	trigger/release; started_at
Pipeline step result	Step attempt in a run	run + step name + attempt no.	step_result_id	pipeline run; started_at
Rejected record	Validation failure for source row/rule	source load + row locator/hash + validation rule	rejected_record_id	source load; validated_at
PO header version	Accepted header observation/version	source + PO number + amendment/version	po_version_id	source load; source effective_at
PO line version	Accepted line observation/version	line identity + amendment/version	po_line_version_id	source load; source effective_at
Delivery schedule version	Accepted schedule observation/version	source + PO + line + schedule key + amendment	schedule_version_id	source load; source effective_at
Supplier commitment	Confirmation observation	source commitment ID, else line/schedule + observed_at + sequence	commitment_id	source load; observed_at
Receipt transaction	Receipt/correction/reversal event	source + receipt document + item/sequence	receipt_txn_id	source load; posted_at
Receipt allocation	Receipt application to schedule/bucket	receipt txn + allocation sequence	receipt_allocation_id	allocation run/event; allocated_at
Inventory snapshot	Product/site observation	source + product + site + snapshot_at	inventory_snapshot_id	source load; snapshot_at
Demand requirement	Dated demand fact	source + requirement ID/version	demand_requirement_id	source load; requirement date/effective_at

Fact-like entity	Exact grain	Natural business key	Generated ID	Primary lineage / effective time
Supplier performance snapshot	Supplier/window/as-of observation	definition version + supplier/site + window + as-of	supplier_perf_snapshot_id	source load/calculation run; as_of_at
Synthetic outcome observation	Hidden outcome for line/site/window	generator version + line/site + outcome window	outcome_observation_id	outcome generation run; outcome_at/window end
Candidate risk evaluation	Eligible line/site/run evaluation	pipeline run + canonical line + site	candidate_evaluation_id	scoring run and all input snapshots; evaluated_at
Candidate contribution	Component result in evaluation	candidate evaluation + component rule ID	contribution_id	evaluation/rule/input ref; evaluated_at
Exception episode	Risk episode sequence for line/site	canonical line + site + episode sequence	episode_id	opening candidate/run; opened_at
Exception event envelope	One recorded episode event	episode + event sequence/correlation event ID	exception_event_id	actor/run/source; effective_at
State event	One state transition	exception event ID	state_event_id	exception event; effective_at
Action	One action/note/plan version	episode + action sequence or client correlation	action_id	actor/source; effective_at
Ownership event	Assignment/delegation change	episode + event sequence	ownership_event_id	mapping/actor; effective_at
SLA obligation	One named clock instance	episode + SLA type + obligation sequence	sla_obligation_id	opening/state/severity event; starts_at
SLA event	One change/condition for obligation	obligation + event sequence/type	sla_event_id	calendar/approval/event; effective_at
Approval request	Controlled change request	episode + request type + client correlation	approval_request_id	requester/event; requested_at
Approval decision	One decision/concurrence	request + required decision role/sequence	approval_decision_id	approver; decided_at
Suppression control	Approved suppression interval	episode + suppression sequence	suppression_id	approval and evidence; starts_at
Evidence reference	One synthetic evidence object/version	evidence source + external reference + version/hash	evidence_id	source/actor; recorded_at
Evidence link	One evidence-to-object relationship	evidence + target type + target ID + purpose	evidence_link_id	linking actor/event; linked_at
Resolution record	One resolution assertion/version	episode + resolution sequence	resolution_id	resolver/evidence/approval; resolved_effective_at
Notification event	One delivery attempt	trigger event + recipient ref + channel + attempt	notification_event_id	template/trigger; created_at
Episode relationship	One directed episode link	from episode + to episode + relationship type	episode_relationship_id	creating event; effective_at

3.3 Append-only, correction and retention matrix

Entity group	Append-only?	Correction/reversal treatment	Minimum portfolio retention
Raw loads and run controls	Yes; run status may progress through controlled states but prior step results remain	New load/run; never replace source object hash	All acceptance runs plus full 24-month reference inputs or reproducible manifests
Master/reference versions	Yes after activation	New effective-dated version; erroneous version is withdrawn with correction link	Entire period needed to interpret retained facts
PO/line/schedule/commitment	Yes at version level	New version with supersedes/corrects relationship; cancellation is a fact, not deletion	Full reference period and any versions referenced by an episode
Receipts/allocations	Yes	Reversal/correction transaction and negative/reversing allocation; original retained	Same as associated PO/episode
Inventory/demand/performance snapshots	Yes	New corrected snapshot/fact linked to prior; reports choose latest valid knowledge as-of	Full reference period and acceptance evidence
Candidate evaluations/contributions	Yes	Never recalculate in place; corrective run creates new evaluation	All scoring/acceptance runs and every evaluation linked to an episode
Episode/events/actions/ownership/SLA	Yes for history; episode current pointers are projections	Correcting event references incorrect event and states corrected values; no deletion/backdating	Entire project life; no automated purge
Approval/suppression/evidence/resolution	Yes	Revoke, expire, withdraw or supersede through new records/events	Entire project life; preserve evidence metadata even if referenced artefact becomes unavailable
Notifications	Yes per attempt	New attempt or cancellation; no overwrite of delivery outcome history	Project life if implemented
Analytics outputs	Rebuildable, not audit authority	Republish a new version; never rewrite a published snapshot silently	Current plus acceptance publications and reconciliation evidence

4. Logical relationship model

4.1 Core procurement cardinalities

- One Source System has zero-to-many Source Loads; each Source Load belongs to exactly one Source System and one Pipeline Run.
- One Purchase Order has one-to-many Purchase-Order Line identities. A line belongs to exactly one order.
- One line has one-to-many line versions; exactly one may be current for a valid-time instant.
- One line has zero-to-many Delivery Schedules. A schedule belongs to exactly one line. A line without schedule detail is allowed only if the line itself provides one accepted requested/expected date and quantity.
- One line has zero-to-many Supplier Commitment Observations. A commitment may optionally reference one schedule; if schedule-specific, it must belong to the same line.
- One line has zero-to-many Receipt Transactions through source line reference. A receipt may allocate to one or more schedules for that same line.
- One Receipt Transaction has zero-to-many Receipt Allocations. Each effective positive receipt must be fully reconciled to schedule allocations plus an explicit unallocated/overreceipt quantity.
- One Product and one Site each have zero-to-many Inventory Snapshots and Demand Requirements. These observations create line exposure through the line's product/site and time horizon; they do not belong exclusively to one PO line.
- One Supplier has zero-to-many Supplier Performance Snapshots; a snapshot may optionally be site-specific.

4.2 Candidate and episode cardinalities

- One successful scoring Pipeline Run has one Candidate Risk Evaluation for every eligible line/site and zero for ineligible lines, with exclusions reconciled separately.
- One Candidate Evaluation has one-to-many Contribution records for all evaluated components, including components marked unavailable. "Missing" must be represented, not omitted.
- A Candidate Evaluation links to zero or one Exception Episode. It links to none when below threshold or data-insufficient; to one new episode when it opens; or to the existing active episode on repeated scoring.
- One Exception Episode is opened by exactly one Candidate Evaluation. One episode then has one-to-many later candidate evaluations across runs.
- One line/site has zero-to-many episodes over time, but at most one non-Closed episode at any instant.

4.3 Workflow cardinalities

- One Episode has one-to-many Exception Event Envelopes and State Events. It must begin with exactly one Open transition and may have at most one transition into Closed.
- One Episode has zero-to-many Actions, Ownership Events, SLA Obligations, SLA Events, Approval Requests, Suppression Controls, Evidence Links, Resolution Records and Notification Events.
- One action may have zero-to-many evidence links and approval requests. Mandatory evidence depends on action category.
- One approval request has one-to-many approval decisions. Most require one manager decision; controlled Critical routes may require a second governance concurrence. Two decisions by the same actor cannot satisfy separation.
- One resolution record has one-to-many evidence links and zero-to-many approval links; a Resolved state event references exactly one valid resolution record.
- A Closed state event references one valid resolution record and closure review evidence/actor.
- One episode may have zero-to-many suppression controls over time, but effective suppression intervals may not overlap.
- One SLA obligation has one-to-many SLA events and zero-to-many approved pause intervals/extensions. A recorded breach is never deleted after extension.

4.4 Predecessor/successor and history

- A material recurrence after Closed creates a new episode and one material-recurrence relationship from prior to successor.
- A successor must reference a Closed predecessor for material recurrence. It cannot reopen or mutate the predecessor.
- Cycles are prohibited. Episode sequence increases monotonically within line/site.
- Supplier and Product identities have one-to-many non-overlapping effective versions. Operational facts reference the durable identity and, where interpretation requires, the version effective at the fact time.
- Historical reporting joins “as known/effective then,” not the current supplier/product description by default. Current-label reporting may be offered separately and labelled.

5. Active-episode uniqueness contract

5.1 Logical key and active definition

The active-episode business key is:

canonical purchase-order-line identity + receiving site identity.

Risk type, schedule identity, risk cause and scoring run are explicitly excluded. “Active” means latest valid state is Open, Assigned, Investigating, Action Agreed, Monitoring, Resolved or Suppressed. Closed is the only inactive state. A mutable `is_active` flag may exist only as a rebuildable projection/cache; it is not the control authority.

5.2 Consistency mechanism

The logical contract requires all episode-opening attempts to pass through one atomic decision boundary:

1. Resolve the source PO/line key to a canonical line identity and receiving site.
2. Validate the candidate and opening rule/version.
3. Acquire transaction-level serialization for the line/site key or rely on an equivalent database-enforced uniqueness boundary.
4. Recheck for a non-Closed episode inside the same transaction.
5. If none exists, create the episode, Open state event, initial SLA obligations and candidate disposition `opened-new-episode` as one atomic unit.
6. If one exists, create no episode; link the evaluation to it with disposition `linked-existing-active-episode` or `suppressed-by-existing-control`.
7. Commit only if both the episode uniqueness constraint and candidate disposition are valid. A losing concurrent transaction re-reads and links to the winner.

The future physical PostgreSQL design must therefore use a database-enforced conditional uniqueness rule over the canonical line/site key for rows without terminal closure, plus transactional opening logic. An application-only pre-check is insufficient.

5.3 Boundary cases

- **Concurrent pipeline runs:** both may retain evaluations, but only one may create the episode. The other links to it. Run order does not decide history; commit serialization does.
- **Repeated scoring:** every evaluation and contribution is retained and linked to the same active episode. It updates the current-risk projection only when its run is eligible and successfully published.
- **Suppressed episode:** remains active and blocks a duplicate. The new candidate records suppression-affected disposition and may trigger recurrence review if control criteria are met.

- **Resolved, not Closed:** remains active and blocks a duplicate. New deterioration records possible-resolution-invalidated evidence and a Reopened/Resolution Withdrawn event, returning the same episode to Investigating without SLA reset.
- **After Closed:** a qualifying material recurrence creates exactly one successor with a new episode sequence and new SLA obligations. The Closed predecessor is untouched.
- **Corrected source key:** the canonical line ID remains stable when a source key was merely wrong. Key-alias history records old and new keys. The correction transaction checks uniqueness for both resolved identities and prevents an accidental merge if each already has an active episode.
- **Receiving-site correction:** if the recorded site was factually wrong and no real supply move occurred, an authorised correction changes the canonical line-site association under a serialized correction process, appends audit evidence and checks the target key for an active collision. It does not create a new episode automatically.
- **Genuine site transfer:** if supply responsibility genuinely moves, the old and new site exposures are distinct business episodes. The old episode remains under human resolution/closure control; a qualifying new-site candidate may open a new episode and link it with relationship site-transfer. Temporary coexistence is permitted because the business keys differ, but reports must avoid presenting it as an unexplained duplicate.

6. Event and history design

6.1 Information classes

Information class	Examples	Update rule
Current-state projection	current state, current owner, current calculated/effective severity, open quantity, current SLA condition	Rebuilt from valid history; may be refreshed or replaced because it is not audit authority
Immutable event history	state, action, ownership, SLA, approval decision, correction, evidence link, notification	Append only; incorrect records are corrected by later linked events
Versioned reference data	supplier/product versions, product-site policy, ownership mapping, rule version, calendar version, UOM conversion	New non-overlapping version; activated versions immutable
Correction or reversal	receipt reversal, source-key correction, corrected snapshot, withdrawn resolution, revoked approval	New record referencing original with reason, actor and effect; original remains visible
Derived reporting output	daily backlog snapshot, duration fact, supplier aggregation, score-performance fact	Reproducibly rebuilt and published under a publication/run ID; never writes back

6.2 Reconstruction rules

- **Current status:** take the latest non-voided state event by effective-at, then recorded sequence. A correcting event never edits the prior transition; it provides an authorised correction and resultant valid state. Prohibit conflicting effective ordering within one episode.
- **Current owner:** start with the latest effective assignment/reassignment. Apply an active approved delegation only for its interval. Ended delegation returns accountability to the retained base owner. Unassignment is explicit.
- **Current risk:** select the latest candidate evaluation for the line/site from a successful scoring publication. Preserve calculated score/severity. Overlay a valid approved priority/severity override as a separate effective value and expiry; do not change the calculated values.
- **Current open quantity:** ordered/base quantity from the valid line version less effective, non-reversed receipt allocations, considering cancellation and permitted tolerance. Schedule residuals sum to line residual.
- **Current SLA:** reconstruct each obligation from original start/due, business calendar, effective pauses, extension decisions, satisfaction and immutable breach events. Never infer “not breached” merely from a revised due time.

For deterministic ordering, every event has a server-recorded monotonic sequence within its aggregate in addition to timestamps. Client-provided timestamps cannot determine audit order alone.

7. Data-quality contract

7.1 Validation outcomes

- **Reject row:** row is excluded; run may continue if control totals and risk safety permit.
- **Quarantine row:** retained for correction and excluded from affected calculations; run may continue with explicit qualification.
- **Stop dataset:** dataset is unusable; no downstream scoring dependent on it.
- **Stop run:** critical coverage, key, referential, chronology or reconciliation failure; no new operational/reporting publication.
- **Warning/qualify:** row accepted but missing/non-critical signal is explicit in scoring and reporting.

Critical failures include ambiguous stable keys, duplicate accepted transactions, unresolved required master references, invalid UOM conversion for quantity exposure, impossible core chronology, missing PO line/product/site/ordered quantity/need date required for eligibility, broken candidate arithmetic, or reconciliation differences. Non-critical failures include optional descriptions, non-essential contact labels, or supplier-performance insufficiency; they become warnings or unavailable signals, never zero risk.

7.2 Dataset contracts

PO header

- Required: source system, PO number, supplier key, order date, order status, currency or explicit not-applicable, source effective time.
- Types: identifiers as non-blank text; dates as ISO-compatible date; effective times timezone-aware; status/currency controlled codes.
- Unique: source system + PO number + amendment/version.
- RI: supplier and source system must resolve.
- Rules: order date cannot be after all line creation dates; cancelled/closed statuses need effective dates.
- Missingness: buyer group may be missing but causes ownership fallback; descriptions optional.
- Policy: duplicate/unknown supplier stops dataset if ambiguous; optional omissions warn.

PO line

- Required: PO reference, source line key/number, product, receiving site, ordered quantity, source UOM, need/requested date, status, effective time.
- Types: quantity finite decimal at product handling precision; dates valid; identifiers text.
- Unique: source PO + line key + amendment/version; canonical current version intervals may not overlap.
- RI: PO, product, site and UOM must resolve.

- Rules: ordered quantity greater than zero unless a cancellation/reversal amendment; open quantity cannot be supplied as authoritative without reconciliation; need date on/after order date unless documented exception.
- Missingness: confirmed/expected date may be absent and becomes a missing signal; product/site/need date are critical.
- Policy: ambiguous key/UOM/product/site stops affected dataset/run; missing confirmation qualifies scoring.

Delivery schedule

- Required: line, schedule key or deterministic source sequence, scheduled quantity, requested/expected date, status.
- Unique: line + schedule key + version.
- RI: schedule line must exist; site must equal line receiving site in the MVP.
- Rules: active schedule quantities reconcile to current line quantity within product precision after cancellations; dates follow order chronology; split dates may differ.
- Missingness: absent schedule dataset is allowed only when line-level quantity/date is sufficient; partial schedule population is critical.
- Policy: reconciliation failure quarantines line and stops its scoring; widespread failure stops run.

Supplier commitment/confirmation

- Required: line or schedule, observed time, committed date or explicit unable-to-confirm code, source/channel.
- Unique: source commitment ID/version, else line/schedule + observed time + sequence.
- RI: referenced schedule belongs to referenced line.
- Rules: committed quantity non-negative and not above outstanding amount without overcommitment code; retroactive observations retain both observed and business-effective time.
- Missingness: entirely missing commitments are accepted as unavailable signals, not safe signals.
- Policy: orphan records quarantine; absent dataset may qualify or stop scoring according to configured freshness criticality.

Receipt

- Required: source receipt identity, line reference, posted/effective time, quantity, UOM, transaction type, source load.
- Unique: source system + receipt document/item/sequence.
- RI: line/product/UOM resolve; schedule reference, if present, belongs to line.
- Rules: positive original quantity; corrections/reversals reference a prior transaction; cumulative effective quantity may exceed ordered only as explicit overreceipt; receipt before order date is critical unless approved migration exception.
- Tolerance: comparisons use product base-UOM handling precision; no unexplained overreceipt is silently clipped.
- Missingness: schedule link may be absent; line link, quantity, UOM and date are critical.

- Policy: duplicates/orphans/invalid UOM quarantine and may stop run if they change an active exposure.

Inventory snapshot

- Required: product, site, snapshot time, on-hand and definition/source; reserved/allocated/available according to selected inventory definition.
- Unique: source + product + site + snapshot time + snapshot version.
- RI: product/site resolve.
- Rules: quantities finite; negative on-hand is allowed only if source semantics permit and is flagged; available must reconcile to documented formula within precision.
- Freshness: age recorded per run; critical staleness threshold is a configuration decision before scoring implementation.
- Missingness: no snapshot for an exposed product/site is data-insufficient or run-critical according to policy, never zero inventory risk.

Demand requirement

- Required: product, site, requirement ID/type, required date, quantity, source, effective time.
- Unique: source + requirement ID + version.
- RI: product/site resolve.
- Rules: positive quantity for original demand; reversal/cancellation references prior; forecast and firm demand types must be distinguishable to prevent double counting.
- Missingness: demand absence is an unavailable signal unless the approved model explicitly defines a complete zero-demand feed.
- Policy: orphan/duplicate demand quarantines; broad feed incompleteness stops exposure scoring.

Supplier performance snapshot

- Required: supplier, measurement window, as-of date, definition version, numerator, denominator, sufficiency flag.
- Unique: supplier + optional site + window + as-of + definition version.
- RI: supplier/site resolve.
- Rules: denominator non-negative; numerator between zero and denominator; rate recomputes exactly; approved minimum observation threshold determines availability. [BRS DQ-009]
- Missingness: insufficient history is accepted with unavailable flag; it cannot contribute safe points.

Master/reference data

- Required: stable key, name/label, active/effective interval, source or approval, version.
- Unique: durable business key; non-overlapping effective versions.
- RI: policy/mapping scopes resolve to product/site/supplier/user dimensions.
- Rules: effective end after start; activation requires approval for rule, calendar, ownership mapping and UOM changes.

- Missingness: optional descriptions accepted; missing key/effective dates reject.

Pipeline/run files

- Required: run ID, dataset type, source, extraction/receipt time, schema version, content hash, source/accepted/rejected counts.
- Unique: run ID; source load object/version within run.
- Rules: source = accepted + rejected/quarantined according to non-overlapping dispositions; eligible = scored; every candidate has disposition; failed run cannot become current publication. [BRS DQ-011, FR-025]
- Policy: any unexplained reconciliation difference stops publication.

7.3 Cross-dataset tolerances

- Quantity equality uses the product base-UOM precision; default portfolio precision is four decimal places unless the product defines a coarser handling increment.
- Rate comparisons retain adequate precision and round only for display; scoring uses stored unrounded values.
- Timestamp equality never merges distinct source events; duplicates require matching stable ID or exact approved deduplication fingerprint.
- Freshness tolerances and minimum supplier-history sample size are rule-catalogue inputs and must be frozen before scoring implementation; the data model stores both observed age/sample and threshold/version.

8. Receipt-allocation contract

8.1 Allocation order

1. Validate and convert the effective receipt quantity to the product base UOM.
2. If a valid schedule link exists, allocate first to that schedule up to its residual open quantity.
3. Allocate any remaining quantity to other open schedules for the same line in ascending expected/committed date, then requested date, then stable schedule sequence.
4. If no schedule link exists, apply the same chronological fallback from the first open schedule.
5. Record quantity beyond total schedule residual as explicit overreceipt/unallocated excess; never force it into a closed schedule or discard it.

8.2 Partial, excess and corrections

- Partial receipts reduce only allocated schedule residuals. A line remains eligible while residual open quantity is greater than product precision and status/date rules permit.
- Multiple receipts accumulate through allocations; cumulative values are derived, not overwritten.
- Overreceipts are valid observations only when explicitly classified. They trigger a DQ/operational review and do not make historical ordered quantity larger.
- A receipt correction/reversal references the original transaction. Its allocation effect reverses original allocations in reverse allocation order unless the source supplies an authoritative corrected schedule allocation. Reversal never edits original allocations.
- Corrected positive replacement quantity is then allocated under the normal rules and linked to both the original and reversal/correction chain.

8.3 UOM assumptions

- Each product has one base inventory UOM per effective product version.
- PO, schedule, receipt, inventory and demand quantities must either use that UOM or a product-specific approved conversion effective at the transaction time.
- Generic cross-product conversions and ambiguous packaging conversions are prohibited in the MVP.
- Fractional quantities are valid only within product handling precision. Missing conversion quarantines the affected record and blocks exposure calculation for the line.

8.4 Residual exposure and workflow effect

Residual line open quantity equals valid ordered/scheduled quantity less effective receipt allocations and valid cancellations, never less than zero for exposure calculation; any negative arithmetic is retained as overreceipt rather than hidden. Projected stock exposure uses residual arrival timing plus inventory and demand observations.

A full effective receipt may create a possible-resolution-evidence event and recalculate risk, but does not automatically transition to Resolved or Closed. A reversal before closure may invalidate resolution and append a Reopened/Resolution Withdrawn event; after Closed it may trigger a linked successor if material recurrence criteria are met. [Operating Model RCV-DEC-001/002, AUT-DEC-001]

9. Candidate-risk data contract

9.1 Required inputs

Every evaluation must identify:

- pipeline run and scoring step;
- canonical PO line and receiving site;
- PO/line/schedule versions used;
- effective receipts and allocation snapshot/cutoff;
- inventory snapshot and freshness;
- demand requirement set/horizon and freshness;
- latest supplier commitment observations used;
- supplier performance snapshot and sufficiency;
- product-site inventory policy and UOM conversion versions;
- rule version and evaluation timestamp;
- any missing, stale, quarantined or qualified signal;
- input-set fingerprint/hash sufficient to prove deterministic reuse.

9.2 Required outputs

- immutable candidate evaluation ID;
- pipeline run, PO line and site;
- rule version and source snapshot/load references;
- eligibility and exclusion reason where applicable;
- one contribution row for every configured component, including unavailable components;
- missing-signal codes and their treatment;
- total risk-priority score from 0 to 100, not labelled probability;
- calculated severity/queue band under the versioned matrix;
- explanation codes and ordered summary reasons;
- candidate disposition;
- zero-or-one linked exception episode;
- evaluated-at and recorded-at instants;
- reconciliation and arithmetic-valid flags.

9.3 Invariants

- Total score equals the approved aggregation of stored contributions and remains within 0-100.
- Same input fingerprint, rule version and configuration produce the same result.
- A missing signal has `available = false`; it is never silently represented as observed zero.

- One evaluation exists per eligible line/site/run. A retry creates a new run or step attempt but cannot create a second accepted evaluation for the same accepted run key.
- Calculated severity is immutable. An approved effective severity override is operational overlay data, not a score edit.
- Candidate disposition and episode link agree: opened/linked/suppressed-by-control require one episode; below-threshold and data-insufficient require none.
- Outcome observations and outcome-generator seed/version are forbidden scoring inputs.

10. Workflow data contract

10.1 Common mandatory event fields

All controlled workflow changes require episode, event ID/type, actor ID/type/role, effective-at, recorded-at, Melbourne-local rendering context, prior/new values or state, reason/source, correlation/causation, and applicable evidence, approval, run, rule and calendar references.

10.2 Stage and control contracts

Activity	Mandatory domain data	Control consequence
Opening	opening candidate/run, line/site, calculated severity/reasons, opened-at, source snapshot refs, dedup outcome, opening rule version	Creates Open event and SLA obligations atomically; automated permitted
Assignment	owner/queue, assignment basis and mapping version, assigning actor/system, assigned-at; manual reason if manual	Open→Assigned only when valid owner exists; prior owner history retained
Investigation	investigator, began-at, triage/cause hypothesis, source facts reviewed, next review/action target	Human Assigned→Investigating; automation cannot assert investigation
Action agreement	action category, rationale, action owner, agreed-at, target/commitment, next action/date, evidence, residual exposure, approvals, monitoring expectation	Human Investigating→Action Agreed; incomplete plan rejected
Monitoring	monitoring start, condition/metric, expected event/date, next review, owner, initiation evidence	Human Action Agreed→Monitoring; condition must be measurable
Resolution	structured category/cause, resolver/time, evidence, outcome quantity/date, monitoring result, residual-risk statement, approval, current score/rule ref, closure-review date	Human transition to Resolved only; unresolved mandatory actions prohibited
Closure	Resolved reference, reviewer/time, closure rationale, evidence/approval, final residual risk/outcome, SLA result, limitation/exceptional-closure flag	Human Resolved→Closed; direct active→Closed rejected
Suppression	prior state/owner, reason/scope, requester/approver, start/expiry/review, evidence, score context, SLA consumed/remaining, recurrence criteria	Human-approved temporary state; no expiry means reject; intervals cannot overlap
Reopening event	invalidating/recurrence reason, triggering observation, previous resolution/suppression, evidence, actor/system, target active state	Pre-Closed: same episode, no SLA reset. Post-Closed: predecessor remains Closed; new successor opens
Ownership change	prior/new owner or queue, reason, authority, mapping/delegation reference, effective time	State unchanged; clocks not reset; base owner retained during delegation
Approval	request type, original/requested values, reason/evidence, requester, required roles, decision(s), conditions, expiry, independence result	No effective change until required approvals complete; reject has zero requested effect
SLA pause	obligation, reason, requested interval, approval, actual start/end, calendar version	Only approved data/system hold or effective suppression pauses; supplier waiting does not

Activity	Mandatory domain data	Control consequence
Escalation	obligation/condition, severity, accountable manager, reason, source event, created time	Immutable event/queue condition; owner and state unchanged
Breach	obligation, original/revised due, breach instant, consumed business minutes, calendar version, cause context	Immutable once recorded; later extension never erases it

11. Audit and evidence contract

11.1 Minimum reconstruction payload

For every controlled change the audit history must answer:

- **Who acted:** actor ID, actor type (human/system), simulated role, delegated/on-behalf-of identity where relevant.
- **What changed:** target aggregate/entity, event type, field/state/value set, previous and new values in typed columns where control-critical plus a canonical diff representation.
- **When:** business-effective instant, server-recorded instant, monotonic event sequence and Melbourne-local display value.
- **Why:** controlled reason code and explanatory note; source/trigger and causation event.
- **Evidence:** zero-to-many evidence references with purpose, availability and integrity metadata; mandatory according to route.
- **Approval:** request and all decisions, conditions, expiry and separation-of-duties result.
- **Version context:** rule version, pipeline run/source load, calendar version, ownership mapping version, application/configuration release.
- **Correction context:** corrected/reversed event or fact, correcting actor, reason, before/after/checksum and downstream effect.

11.2 Audit safeguards

- Standard application and reporting roles cannot update/delete historical event rows.
- Administrative corrections use a distinct attributable path and append correction events. [BRS GOV-010]
- Event IDs and sequences are server-generated; client correlation IDs support idempotency but are not trusted as primary identity.
- Before/after values must be minimised. Secrets and unnecessary personal data are excluded; evidence content is not copied into audit payloads.
- Hashes demonstrate byte/value consistency within the portfolio but do not prove legal immutability or tamper-proof storage.
- An audit-completeness projection reports missing mandatory fields/links by event type. It does not “repair” incomplete history.

12. Reporting model recommendation

The analytics layer should use curated, read-only facts and conformed dimensions tied to a successful publication. It should not expose raw operational tables directly to Power BI.

Analytical need	Recommended fact/projection	Core dimensions/measures
Current exception backlog	Current episode snapshot at latest successful publication	episode, state, calculated/effective severity, owner/unassigned, site, supplier, product, age, next action, breach flags
Lifecycle duration	State interval fact derived from state events	episode, from/to state, entry/exit, business/calendar minutes, severity at entry, site/owner
SLA performance	SLA obligation/event fact	SLA type, original/revised due, business minutes, pause minutes, met/breached, escalation, severity, owner/site
Supplier risk	Candidate/episode/supplier outcome facts	supplier/site/product, candidate count, episode count, contribution reasons, repeat episodes, OTIF sufficiency, outcomes
False-positive analysis	Candidate evaluation joined only in analytics to independent outcome observation	rule version, threshold/band, outcome window, TP/FP/FN/TN disposition, exclusion, lead time
Risk-score performance	Candidate evaluation and contribution facts	score/band, rule version, reasons, outcome, useful lead time, workload, missing-signal cohort
Pipeline health	Pipeline run/source-load/step fact	status, duration, counts, freshness, rejects, reconciliation, publication age
Audit completeness	Controlled-event compliance fact	event type, mandatory-field/link checks, actor/approval/evidence completeness, correction count

Conformed dimensions should include Date, Melbourne Business Date, Site, Supplier (type 2), Product (type 2), User/Role, State, Severity, Rule Version, Calendar Version, Candidate Disposition, Explanation/Reason, Resolution Category, Action Category, SLA Type and Pipeline Run. Keep current calculated severity and effective overridden severity as separate columns/measures.

Publication uses an immutable publication ID and source pipeline-run set. A single controlled pointer identifies the latest successful publication. Failed runs remain visible in pipeline health but cannot replace the current operational/reporting dataset. [BRS FR-025, REP-007 to REP-009]

13. Privacy and security classification

Classification	Examples	Handling contract
Public portfolio-safe	synthetic site/supplier/product IDs and names; synthetic PO/receipt/demand data; rule labels; aggregate metrics; documented assumptions	May enter repository and demo after synthetic-data and limitation checks
Internal operational	detailed synthetic transaction rows, exact ownership mappings, workflow notes, run diagnostics, source row locators	Keep access role-limited in design; publish only what aids the portfolio
Personal or identity-related	simulated user display name, user ID, role, email-like recipient reference	Use fictional identities; minimise; do not infer real identity; separate from public exports where unnecessary
Sensitive governance data	approval decisions, override reasons, accepted residual risk, correction history, audit gaps, evidence hashes/links, admin actions	Restrict write/read by role; retain complete audit history; redact narrative details from public screenshots if needed
Secrets—never in database or repository	passwords, API keys, tokens, connection strings with credentials, private keys, OAuth refresh tokens, real confidential evidence, real personal/contact data	External secret mechanism only; use placeholders in configuration; secret scan before release [BRS SEC-004]

Database usernames and non-secret connection metadata may be configuration, but credentials are never operational records. Evidence references must be synthetic and resolvable without embedding signed URLs, tokens or document contents.

14. MVP simplification review

14.1 Must exist

- source system/load, pipeline run/step and rejected-record controls;
- site, supplier, product, product-site policy, user, ownership mapping, rule/calendar version;
- PO, stable line identity/version, schedule, receipt and receipt allocation;
- inventory, demand, supplier commitment and performance observations;
- candidate evaluation and contribution;
- episode, event envelope/state events, action, ownership, SLA obligation/events;
- approval request/decision, suppression, evidence metadata/link and resolution;
- episode relationship for recurrence;
- curated reporting publication and reconciliation controls.

14.2 Generic structures suitable for MVP

- Pipeline Step Result can represent ingestion, validation, scoring and publication steps.
- Exception Event Envelope can provide shared audit metadata for typed state, ownership, SLA and correction events.
- Evidence Link can associate one evidence reference with several controlled objects.
- Episode Relationship can cover recurrence and site transfer without separate predecessor/successor tables.
- A controlled reason-code catalogue can serve event, rejection, explanation and correction domains with namespaces; do not collapse their meanings into ungoverned free text.

14.3 May be simulated

Role switching, duty queue, delegation, two-person approvals, Victorian holiday fixtures, evidence references, concurrent opening collisions, receipt reversals, notification attempts, and synthetic outcome generation may be fixtures. Simulation must be labelled and cannot substantiate real authentication, separation of duties or operational performance.

14.4 Defer

- live ERP/WMS/EDI integration and change-data capture;
- document/binary evidence repository and legal records management;
- configurable workflow/rule/calendar designer;
- multi-jurisdiction calendars and 24/7 rosters;
- supplier portal, email/Teams delivery and retry orchestration;
- row-level enterprise security, SSO and production IAM;
- workload optimisation, multiple concurrent owners and schedule-line case management;
- WORM storage, disaster recovery and legal retention automation;

- AI recommendations and all AI storage until P1-P4 gates pass.

14.5 Avoidable over-engineering

- A separate table for every event type when a shared envelope plus a few typed child structures is sufficient.
- Bitemporal modelling on every descriptive field. Use valid-time versions plus recorded-at/correction history only where business interpretation requires it.
- Treating risk causes or delivery schedules as exception keys.
- A mutable “current score” row that overwrites evaluations.
- General-purpose JSON for control-critical values. JSON/diff payloads may supplement, not replace, typed state, owner, score, quantity, approval and due-time fields.
- Real-time streaming, microservices, event brokers or data lake architecture for a daily 50,000-line portfolio workload.

15. Frozen modelling decision register

DAT-DEC-001 — Layered authority

Decision: Separate raw, validation, master, operational, scoring, workflow, governance, pipeline and analytics authorities.

Rationale: Prevents a report, score or raw extract from silently becoming operational truth.

Alternatives rejected: Single wide operational table; BI directly on raw files.

Affected requirements: FR-001 to FR-004, FR-025, DQ-011, REP-007 to REP-009.

Implementation consequence: Physical schemas/views must preserve layer boundaries and read/write roles.

Test implication: Lineage and reconciliation tests cross every layer; failed run cannot publish.

Remaining uncertainty: Final Power BI connection path (views versus controlled extract).

DAT-DEC-002 — Stable identities plus versions

Decision: Separate durable supplier, product and PO-line identities from effective-dated versions and source-key aliases.

Rationale: Amendments and corrected keys must not rewrite score/episode history.

Alternatives rejected: Mutable master rows; source natural keys as all foreign keys.

Affected requirements: DQ-001, DQ-002, GOV-003, GOV-010.

Implementation consequence: Generated IDs, version intervals and alias/correction controls are required.

Test implication: Key correction and as-of-history fixtures must preserve old references.

Remaining uncertainty: Exact synthetic source amendment semantics.

KEY-DEC-001 — Episode key

Decision: Active uniqueness uses canonical PO-line identity plus receiving site; risk type, schedule and score run are excluded.

Rationale: Matches the frozen operational grain and prevents duplicate work.

Alternatives rejected: Episode per risk cause, schedule, candidate or run.

Affected requirements: FR-011, FR-012, RULE-013/014, Operating Model EPI-DEC-001/002 and DUP-DEC-001.

Implementation consequence: Conditional database uniqueness and serialized opening are mandatory.

Test implication: Repeated and concurrent candidates create one active episode and multiple evaluations.

Remaining uncertainty: Whether real source systems ever allow independent schedule ownership/sites.

KEY-DEC-002 — Active includes Resolved and Suppressed

Decision: Every state except Closed is active for duplicate prevention.

Rationale: Resolved awaits closure review; Suppressed is dormant, not complete.

Alternatives rejected: Mutable active flag; treating Resolved/Suppressed as inactive.

Affected requirements: LIF-008/009, Operating Model LIF-DEC-004/005.

Implementation consequence: Current-state projection and uniqueness predicate must agree.

Test implication: New candidates link to Resolved/Suppressed episodes instead of opening duplicates.

Remaining uncertainty: None for MVP.

KEY-DEC-003 — Site corrections versus real transfer

Decision: Factual site correction preserves canonical episode under controlled collision checks; a genuine transfer may create a linked new-site episode.

Rationale: Data correction is not a new risk, while operational relocation is.

Alternatives rejected: Always mutate site; always open successor.

Affected requirements: GOV-010, DQ-001/002, frozen stable line/site grain.

Implementation consequence: Site-change reason and episode relationship type are required.

Test implication: Collision, correction and genuine-transfer scenarios need separate fixtures.

Remaining uncertainty: Source field that distinguishes correction from operational transfer.

EVT-DEC-001 — Event envelope with typed details

Decision: Use one immutable event envelope for shared audit fields and typed child records for state, ownership, SLA and other control-critical events.

Rationale: Consistent audit reconstruction without untyped event blobs.

Alternatives rejected: Separate duplicated audit columns everywhere; JSON-only event store.

Affected requirements: FR-019, FR-024, GOV-001/002, LIF-012.

Implementation consequence: Event ID/correlation/sequence is shared; domain fields remain constrained.

Test implication: Mandatory-envelope test plus type-specific contract tests.

Remaining uncertainty: Which low-risk action/note event types can share one typed structure.

EVT-DEC-002 — Current state is a projection

Decision: Current status, owner, risk and SLA condition are rebuilt from history; cached pointers are non-authoritative.

Rationale: Avoids silent loss while supporting fast queue reads.

Alternatives rejected: Snapshot-only row; UI history log detached from state.

Affected requirements: BR-005, GOV-001 to GOV-003, FR-013 to FR-024.

Implementation consequence: Deterministic ordering and projection rebuild/reconciliation are required.

Test implication: Delete/rebuild projections and prove equality with golden episode.

Remaining uncertainty: Projection refresh strategy under expected concurrency.

EVT-DEC-003 — Reopened is an event

Decision: Pre-Closed reopening returns the same episode to an active work state; post-Closed recurrence creates a successor. No same-episode SLA reset.

Rationale: Preserves closure and response metrics.

Alternatives rejected: Reopened state; mutating Closed; resetting old clock.

Affected requirements: BRS LIF-010 as superseded; Operating Model LIF-DEC-004, SLA-DEC-005, RCV-DEC-002.

Implementation consequence: Reopen reason/trigger and predecessor relationship required.

Test implication: Before/after closure reversal scenarios; old breach remains.

Remaining uncertainty: Material recurrence threshold belongs to rule catalogue.

RCV-DEC-001 — Explicit receipt allocations

Decision: Persist receipt-to-schedule allocations and explicit excess; use chronological fallback when schedule link is missing.

Rationale: Makes residual exposure reproducible across split schedules and reversals.

Alternatives rejected: Cumulative received quantity on line; equal distribution.

Affected requirements: DQ-003/008, RULE-014, FR-021, Operating Model RCV-DEC-001/002.

Implementation consequence: Receipt allocation is a required entity.

Test implication: Partial, multiple, overreceipt, correction and reversal fixtures.

Remaining uncertainty: Source-specific allocation hints beyond schedule ID.

RCV-DEC-002 — No automatic resolution from receipt

Decision: Objective receipts create possible-resolution evidence only; a human controls Resolved and Closed.

Rationale: Corrections/reversals can invalidate apparent completion.

Alternatives rejected: BRS LIF-007 automatic resolution path.

Affected requirements: FR-021/022, Operating Model LIF-DEC-003 and AUT-DEC-001.

Implementation consequence: Evidence event and human resolution record are distinct.

Test implication: Full receipt leaves state unchanged until authorised human action.

Remaining uncertainty: Future controlled automation is explicitly deferred.

DQ-DEC-001 — Missing is not safe

Decision: Missing/stale signals are explicit availability states and cannot become zero-risk observations.

Rationale: Prevents false confidence.

Alternatives rejected: Null-to-zero defaults; dropping unavailable contributions.

Affected requirements: DQ-005 to DQ-009, FR-007.

Implementation consequence: Candidate contributions include availability/missing codes.

Test implication: Missing-signal fixtures reproduce score qualification and explanation.

Remaining uncertainty: Stop-versus-qualify freshness thresholds.

DQ-DEC-002 — Reconciliation gates publication

Decision: Any unexplained source-to-publication count difference blocks publication.

Rationale: Every row requires one non-overlapping disposition.

Alternatives rejected: Warning-only mismatch; dashboard on partially accepted failed run.

Affected requirements: DQ-011, FR-003/004/025, REP-007/009.

Implementation consequence: Control totals at source, accepted, rejected, eligible, scored, candidate, episode and published stages.

Test implication: Injected mismatch keeps prior successful publication current.

Remaining uncertainty: Dataset-specific acceptable warning thresholds do not relax arithmetic reconciliation.

DQ-DEC-003 — Product-specific base UOM

Decision: Compare quantities only in product base UOM using effective approved conversions and handling precision.

Rationale: Generic conversion can distort exposure.

Alternatives rejected: Unit labels only; global pack conversions.

Affected requirements: DQ-003, DQ-008.

Implementation consequence: Product version, UOM conversion and converted/source quantities retained.

Test implication: Valid, missing, ambiguous and effective-date conversion boundaries.

Remaining uncertainty: Synthetic product handling increments.

DQ-DEC-004 — Outcome independence

Decision: Synthetic outcomes use separate generator/version/seed lineage and are prohibited scoring inputs.

Rationale: Prevents circular performance evidence.

Alternatives rejected: Deriving outcomes from score thresholds or triggered rules.

Affected requirements: DQ-010, REP-006.

Implementation consequence: Separate outcome observation and restricted scoring access path.

Test implication: Changing score thresholds cannot regenerate outcomes; leakage test required.

Remaining uncertainty: Final outcome-window definitions.

SLA-DEC-DAT-001 — Obligations separate from events

Decision: Store original SLA obligation separately from pauses, extensions, breaches and satisfaction events.

Rationale: Revised due dates must not erase original performance evidence.

Alternatives rejected: Mutable due date/breach flag on episode.

Affected requirements: SLA-002 to SLA-009; Operating Model SLA-DEC-001 to 006.

Implementation consequence: One-to-many obligation/event model with calendar version.

Test implication: DST, holiday, pause, reassignment, extension and breach persistence tests.

Remaining uncertainty: Closure-review target and maximum prospective extension.

APR-DEC-DAT-001 — Request and decisions separated

Decision: One approval request supports one or more role-specific decisions/concurrences.

Rationale: Critical routes may require two independent authorities.

Alternatives rejected: Approver columns on action; single combined approval row.

Affected requirements: GOV-004/005, SLA-008/009, Operating Model APR-DEC-001 to 003.

Implementation consequence: Required decision set and independence validation.

Test implication: Self-approval, same-person concurrence, rejection, conditional and expiry cases.

Remaining uncertainty: Exact routes needing governance concurrence.

AUD-DEC-001 — Corrections append

Decision: Historical fact/event correction appends a linked correction/reversal with before/after, reason and actor.

Rationale: Audit history must survive mistakes and administrative actions.

Alternatives rejected: UPDATE/DELETE of history; backdated silent replacement.

Affected requirements: GOV-001/002/010, BR-005.

Implementation consequence: Correction links and privileged attributable correction path.

Test implication: Standard role mutation fails; admin correction adds history and changes projection only.

Remaining uncertainty: Database-superuser controls cannot be proven in the portfolio.

AUD-DEC-002 — Evidence metadata, not binary storage

Decision: Store synthetic resolvable evidence metadata and links; defer document content storage.

Rationale: Meets portfolio audit demonstration without confidential-document/security scope.

Alternatives rejected: Binary documents in database; public real-world links.

Affected requirements: GOV-007, SEC-001, FR-018/021/022.

Implementation consequence: Evidence reference/link and availability/hash checks.

Test implication: Mandatory evidence, broken reference and corrected evidence scenarios.

Remaining uncertainty: Final mandatory evidence types by action/resolution category.

ANA-DEC-001 — Successful-publication pointer

Decision: Reporting uses an immutable publication and controlled latest-success pointer; failed runs remain diagnostic only.

Rationale: Prevents stale/failed data from masquerading as current.

Alternatives rejected: Reporting latest run regardless of status; direct mutable extracts.

Affected requirements: FR-025, REP-001 to REP-010.

Implementation consequence: Publication manifest, run set and reconciliation checksum.

Test implication: Failed-run scenario retains previous current dataset and exposes age/failure.

Remaining uncertainty: PostgreSQL views versus CSV/Parquet delivery.

SEC-DEC-001 — Synthetic identity and secrets boundary

Decision: Database contains only fictional identities and no credentials, tokens, real evidence or real personal data.

Rationale: Public portfolio has no authority to process real operational data.

Alternatives rejected: Masked real data; credentials in configuration tables.

Affected requirements: SEC-001 to SEC-005, GOV-008.

Implementation consequence: Classification metadata, sample placeholders and role-separated database design.

Test implication: Synthetic-data and secret scans; negative database-role tests.

Remaining uncertainty: Hosted demo remains gated/deferred.

Appendix A. Recommended logical entity list

Source/control: Source System; Source Load; Pipeline Run; Pipeline Step Result; Rejected Record; Reconciliation Result; Analytics Publication.

Master/reference: Site; Supplier; Supplier Version; Product; Product Version; Product-Site Inventory Policy; User; Ownership Mapping; Rule Version; Rule Component Definition; Calendar Version; UOM Conversion; controlled code catalogues.

Transactions/observations: Purchase Order; PO Header Version; PO-Line Identity; PO-Line Key Alias; PO-Line Version; Delivery Schedule/Version; Supplier Commitment Observation; Receipt Transaction; Receipt Allocation; Inventory Snapshot; Demand Requirement; Supplier Performance Snapshot; Synthetic Outcome Observation.

Scoring: Candidate Risk Evaluation; Candidate Risk Contribution.

Workflow/governance: Exception Episode; Exception Event Envelope; State Event; Action; Ownership Event; SLA Obligation; SLA Event; Approval Request; Approval Decision; Suppression Control; Evidence Reference; Evidence Link; Resolution Record; Notification Event; Episode Relationship.

Derived only: Current Episode Projection; Current Risk Projection; Current Owner Projection; Current SLA Projection; curated analytical facts/dimensions.

Appendix B. Entity grain matrix

The authoritative grain, natural key, generated ID, lineage/effective time, append-only treatment and retention are frozen in Sections 3.2 and 3.3. Physical design must not combine entities with different grains merely to reduce table count. In particular, Candidate Evaluation, Contribution, Episode, State Event, Receipt and Receipt Allocation remain distinct.

Appendix C. Relationship and cardinality summary

Parent	Child	Cardinality	Mandatory/optional rule
Purchase Order	PO Line	1 to many	Order requires at least one line
PO Line	Delivery Schedule	1 to zero/many	Zero allowed only with complete line-level date/quantity
PO Line	Receipt Transaction	1 to zero/many	Receipt requires one canonical line
Receipt	Receipt Allocation	1 to zero/many	Positive accepted receipt reconciles allocations plus explicit excess
Product/Site	Inventory Snapshot	1 to zero/many	Required signal policy determines scoring gate
Product/Site	Demand Requirement	1 to zero/many	Absence is unavailable unless feed is proven complete
Pipeline Run	Candidate Evaluation	1 to zero/many	Successful scoring run has one per eligible line/site
Candidate Evaluation	Contribution	1 to many	All configured components represented
Candidate Evaluation	Episode	many to zero/one	Episode opened or linked only when disposition requires
Episode	State/Event/Action/Owner/SLA/Approval/Evidence/Resolution	1 to many or zero/many	Episode requires Open state event; others route-dependent
Approval Request	Approval Decision	1 to many	Complete required set before requested effect
Episode	Successor Episode	1 to zero/many	Directed acyclic relationship; recurrence predecessor Closed
Supplier/Product	Version	1 to many	Effective intervals do not overlap

Appendix D. Active-episode uniqueness contract

The complete contract is Section 5. Physical acceptance requires: database-enforced conditional uniqueness; atomic opening; concurrency loser links its evaluation to winner; Resolved and Suppressed block duplicates; Closed permits exactly one qualifying successor per recurrence trigger/correlation; key/site corrections perform collision checks; and a mutable active flag cannot be the sole control.

Appendix E. Data-quality contract

The complete incoming-data contract is Section 7 and the receipt-specific contract is Section 8. Before build, each dataset must receive a machine-readable companion specification using the same required fields, types, uniqueness, RI, chronology, tolerance, missingness and failure severity defined here. This document intentionally avoids executable schema or validation code.

Appendix F. Frozen modelling decision register

The frozen register is Section 15. New decisions must receive a new ID and must not silently edit rationale or rejected alternatives. A change affecting episode identity, active uniqueness, human control, event immutability, receipt allocation, outcome independence or time semantics requires a new document version and regression-test impact review.

Appendix G. Questions that must be answered before physical PostgreSQL implementation

1. What exact source fields distinguish a factual receiving-site correction from a genuine operational site transfer?
2. What opening and material-recurrence thresholds/codes are frozen in the risk-rule catalogue?
3. What freshness threshold stops versus qualifies scoring for PO, receipt, inventory, demand, commitment and supplier-performance feeds?
4. What minimum supplier-performance observation count and measurement window are approved?
5. Which inventory definition is authoritative: on-hand minus reservations, supplied available balance, or another documented formula?
6. How are forecast and firm demand deduplicated or netted to prevent double counting?
7. Are line-level requested date and expected date sufficient when no delivery schedules exist, and can schedules ever have different receiving sites in synthetic data?
8. What product handling precisions and UOM conversions will the synthetic catalogue contain?
9. Which resolution categories and action categories require which minimum evidence types and monitoring periods?
10. Which approval routes require governance concurrence, and what maximum suppression/extension durations apply by severity?
11. What is the closure-review target and does it create a formal SLA obligation or a non-SLA expectation?
12. Which active state should a suppression expiry or withdrawn resolution return to for each prior-state scenario?
13. What constitutes the immutable input-set fingerprint for deterministic scoring and rerun comparison?
14. What is the idempotency/correlation key supplied by the future UI for retries of actions and transitions?
15. Will Power BI consume curated PostgreSQL views or controlled CSV/Parquet extracts for the final demonstration?
16. What simultaneous writer/concurrent-run limit will be tested and disclosed for the portfolio?
17. Will all timestamps originate at the application/database server, and how will erroneous source local timestamps declare timezone/offset?
18. What portfolio retention/archive policy applies beyond “no automated purge” and the required 24-month evidence window?
19. Which current-state projections will be persisted versus exposed as views, and what rebuild/reconciliation SLA applies?
20. Is notification storage omitted entirely from the first schema migration or included dormant for later use?

Appendix H. Critical self-review

What is strong

- The model preserves the candidate/episode separation and frozen lifecycle semantics.
- Active uniqueness is defined as a concurrency and identity contract, not a flag.
- Receipt allocation, reversal and UOM treatment make residual exposure reproducible.
- Audit reconstruction spans source, run, rule, calendar, actor, evidence and approval context.
- Synthetic outcomes are structurally separated from risk scoring, supporting credible recall/false-positive testing.
- Reporting currency is protected from failed runs.

Material limitations and risks

- The model is broad for one developer. Event envelope, versions and analytics projections add justified complexity, but implementation should stop at the MVP list before notifications, hosting or AI.
- “Immutable” is an application/database-role design goal, not proof against a database superuser, storage administrator or compromised host. No WORM or independent audit store is claimed.
- The active uniqueness rule is only as reliable as canonical line/site resolution. Ambiguous source-key correction remains a high-risk implementation area.
- Genuine site transfer semantics are designed but not validated with procurement practitioners or a real ERP.
- Business-calendar targets, suppression limits, recurrence thresholds, supplier-history sufficiency and freshness thresholds remain portfolio assumptions until Appendix G is resolved.
- Effective-time corrections can become complex. The MVP should prohibit arbitrary backdating and require explicit administrative correction events rather than attempt full bitemporal reconstruction for every field.
- Generic event/evidence links can weaken referential enforcement if implemented carelessly. Physical design should constrain target types and use typed foreign keys for control-critical links where practical.
- Current projections can drift from history. Rebuild-and-reconcile tests are mandatory; queue performance must not justify making projections the audit authority.
- Synthetic-data scale can prove deterministic behaviour and portfolio performance only under the test environment. It cannot prove enterprise throughput, security, operational benefit or user adoption.

Final assurance statement

This logical model is sufficiently defined to begin physical PostgreSQL design **after** the high-impact questions in Appendix G are resolved or explicitly frozen as assumptions. It is not production-ready, has not been validated by representative procurement users, and does not substantiate enterprise security, legal retention, tamper-proof auditability or realised business benefit.

Governing sources

Supply Chain Exception Control System — Project Charter v1.0 (19 July 2026).

Supply Chain Exception Control System — Business Requirements Specification v1.0 (19 July 2026).

Supply Chain Exception Control System — Frozen Operational Exception Lifecycle and Operating Model v1.0 (20 July 2026).